

ASP3231 Assignment 2

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Question 1

1a.

The ideal diffraction limited solution refers to the maximum theoretical resolution a lens can achieve, assuming there are no lens imperfections. The only resolution arises from the physical diffraction itself. The standard approximation defines the angular resolution as

$$\theta \approx \frac{\lambda}{D}. \quad (1)$$

Given a wavelength of 2.15 microns and a 10.5 metre diameter, $\theta = 2.04762 \cdot 10^{-7}$ radians, or 42.24 milliarcseconds. Given the Keck telescope is only about 30% larger than the ideal limit, this indicates that some atmospheric blurring or other imperfections are present on the telescope.

The Strehl ratio is a percentage from 0 to 1 that defines how 'perfect' the peak intensity of the point spread function is. This refers to how the telescope responds to a single point of light, and it can be affected by atmospheric turbulence and impurities. The Strehl ratio of 0.4 indicates it is 40% of the ideal system (which sits at 1.0), which is still considered very good for ground-based telescopes.

1b.

Typical natural guide star systems require the guide star to measure the overall image motion (tip tilt) as well as the aberrations of the light. To measure these aberrations, the light is split up into hundreds of tiny apertures, meaning the guide star needs to be much brighter otherwise it would essentially receive no light. The Keck LGS AO system, however, uses an artificial laser to plant a bright star high in the earth's atmosphere, which is used to determine these aberrations. The natural guide star, therefore, is only used to measure the tip-tilt, meaning it can be far fainter than typical as its light is being captured by the full sensor.

1c.

The Fried length r_0 is defined to be the typical size effective at a wavelength of $0.5\mu\text{m}$ (Rieke 2012). Since we want to find the isoplanatic angle at $2.15\mu\text{m}$, we need to scale the length. Since $r_0 \propto \lambda^{6/5}$ (Rieke 2012),

$$r_0 = r_0|_{0.5\mu\text{m}} \cdot \left(\frac{2.15}{0.5}\right)^{6/5} = 1.44\text{m},$$

using the given values.

The isoplanatic angle, applying the small angle approximation (Rieke 2012) is given by

$$\theta_0 = 0.31 \frac{r_0}{h}. \quad (2)$$

Using the scaled r_0 value and the altitude of 1800m, the isoplanatic angle is $2.479 \cdot 10^{-4}$ radians, or 51.12 arcseconds.

1d.

The main difference is that JWST is in space, and the Keck LGSAO telescope is on the ground. Earth's atmosphere absorbs most of the incoming infrared light, including the high redshift light, and combined with the infrared emission of the atmosphere and the telescope itself, makes it extremely difficult to obtain spectra of highly redshifted galaxies.

Question 2

2a.

These are X-rays.

2b.

To convert between keV and erg, we can divide the values by $6.242 \cdot 10^8$. The energy range is between $0.801 - 6.41 \cdot 10^{-9}$ erg, and hence the typical energy (average) is $3.6 \cdot 10^{-9}$ erg.

2c.

The total energy from the absorbed photons will be the number of photon counts (124) multiplied by the energy per photon ($3.6 \cdot 10^{-9}$ erg).

$$E_{\text{total}} = 4.46 \cdot 10^{-7} \text{ erg.}$$

The flux is the energy per unit area per second, or

$$F = \frac{E_{\text{total}}}{A_{\text{eff}} \times t}. \quad (3)$$

Using an effective area of 1.5 cm^2 and an exposure time of 100 seconds, this evaluates to $2.98 \cdot 10^{-9} \text{ erg cm}^{-2} \text{ s}^{-1}$.

2d.

Over the 100 second exposure at 0.3 counts per second, there were 30 background counts. So the total count is 154 counts. The count uncertainty will be

$$\sigma_{\text{counts}} = \sqrt{N} \approx 12.4 \text{ counts,}$$

resulting in a fractional uncertainty of $\frac{12.4}{124} \approx 0.1$.

The $1\text{-}\sigma$ uncertainty in the flux is the flux multiplied by this fractional uncertainty, $\approx 2.98 \cdot 10^{-10} \text{ erg cm}^{-2} \text{ s}^{-1}$.

2e.

The spectral flux density is the total flux divided by the range

Question 3

3a.

3b.

3c.

1 References

Rieke, G. H. (2012). *Measuring the Universe: A Multiwavelength Perspective*. eng. 1st ed. Cambridge: Cambridge University Press. ISBN: 9780521762298.